Exps 9–10 · Flip-flops from NAND gates; 7474 and 7473; flip-flop conversion

Manual pp. 4–8 (second numbering) · ICs 7400, 7410, 7474, 7473 · use the trainer's debounced pulser as the clock

MANUAL EXPS 9–10

Study SR, JK and T flip-flops built from NAND gates, the 7474 (positive-edge D, preset and clear) and the 7473 (negative-edge JK, clear), and conversions SR → D, D → SR. Viva form: "Your NAND JK flip-flop with J = K = 1 ends in a random state. Why, and how do the ICs avoid it?"

GUESS FIRST

The gate JK is *level*-triggered: it reacts for as long as CLK = 1. What happens if Q flips while CLK is still 1?

Exp 9 · From gates

Clocked SR (Fig 9.1), one 7400:

- G1 = NAND(S, CLK): 1, 2 → 3.
- G2 = NAND(R, CLK): 4, 5 → 6.
- G3 = NAND(G1, Q'): 3 → 9, 11 → 10 → **8 = Q**.
- G4 = NAND(G2, Q): 6 → 12, 8 → 13 → **11 = Q'**.

CLK	S R	Q_{n+1}
0	x x	Q_n (hold)
1	0 0	Q_n
1	0 1	0
1	1 0	1
1	1 1	**invalid** (Q = Q' = 1, then unpredictable when CLK falls)

JK (Fig 9.2): feed Q' back into the J gate and Q into the K gate. That makes S = R = 1 impossible, and J = K = 1 toggles instead.

- 7410: J → 1, CLK → 2, Q' → 13 → 12; K → 3, CLK → 4, Q → 5 → 6.
- 7400 latch: 7410-12 → 1, Q' → 2 → **3 = Q**; 7410-6 → 4, Q → 5 → **6 = Q'**.

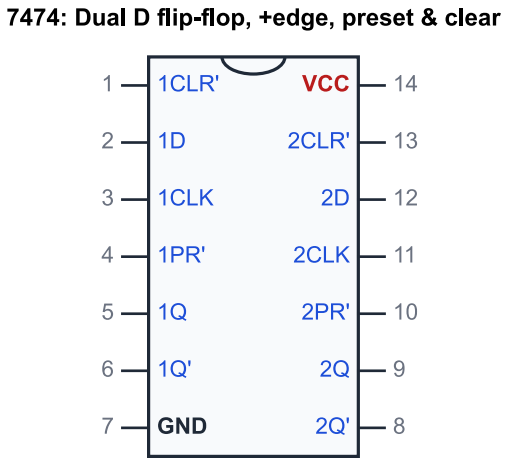
CLK pulse	J K	Q_{n+1}
↑ rising	0 0	Q_n
↑ rising	0 1	0
↑ rising	1 0	1
↑ rising	1 1	Q_n' (toggle)

RACE-AROUND

This JK is level-triggered. With J = K = 1, it toggles, the new Q feeds back, and it toggles again, as many times as fit inside one CLK-high time. The final state on the trainer is random. Edge-triggered ICs (7473, 7474) and master-slave designs act only at a clock **edge**, so each pulse gives exactly one toggle.

T flip-flop (Fig 9.3): tie J = K = T. T = 0 holds; T = 1 toggles on every clock.

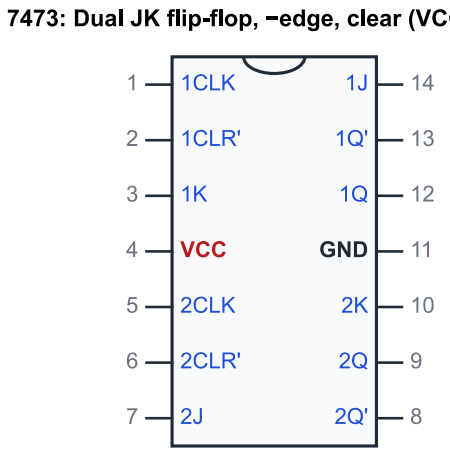
Exp 10 · Flip-flop ICs



7474: two positive-edge D flip-flops with active-LOW preset and clear.

7474 (FF 1): CLR' 1, D 2, CLK 3, PR' 4, Q 5, Q' 6.

PR'	CLR'	CLK	D	Q Q'
0	1	x	x	1 0 (preset, ignores clock)
1	0	x	x	0 1 (clear)
0	0	x	x	1 1 (not allowed)
1	1	↑ rising	1	1 0
1	1	↑ rising	0	0 1
1	1	0	x	Q_n Q_n'



7473: two negative-edge JK flip-flops with active-LOW clear. VCC = pin 4, GND = pin 11.

7473 (FF 1): CLK 1, CLR' 2, K 3, J 14, Q 12, Q' 13. **Table 10.2, filled:**

CLR'	CLK	J K	Q_{n+1}	Q_{n+1}'
0	x	x x	0	1
1	↓ falling	0 0	Q_n	Q_n'
1	↓ falling	1 0	1	0
1	↓ falling	0 1	0	1
1	↓ falling	1 1	Q_n'	Q_n (toggle)
1	1 (no edge)	x x	Q_n	Q_n'

Conversions:

Want	Using	Logic
D from SR (Fig 10.1)	S = D, R = D'	7404 inverts D for R
SR from D (Fig 10.2)	D = S + R'Q	7404 (R'), 7408 (R'·Q), 7432 (+S) into the 7474's D; Q fed back
JK from T (exercise)	T = JQ' + KQ	
T from D (exercise)	D = T ⊕ Q	one 7486
SR from JK (exercise)	J = S, K = R	S = R = 1 must still be avoided
SR from NOR (exercise)	cross-coupled NORs	active-HIGH inputs; S = R = 1 forbidden

WRITE IN YOUR RECORD

Result: the truth tables of the NAND-gate SR, JK and T flip-flops and of the 7474 (positive-edge D, PR'/CLR') and 7473 (negative-edge JK, CLR') were verified. SR → D (S = D, R = D') and D → SR (D = S + R'Q) conversions were verified.

Theory for the record: a flip-flop is a 1-bit memory element. The clocked SR flip-flop has an invalid state for S = R = 1. The JK removes it by feedback, giving toggle for J = K = 1. A level-triggered JK suffers race-around, which edge triggering or master-slave construction prevents. The D flip-flop copies D at the clock edge; the T flip-flop toggles when T = 1. Preset and clear are asynchronous, active-LOW inputs that override the clock. Conversion uses the excitation table of the available flip-flop to find its inputs in terms of the desired inputs and Q.

CHECK 1

(a) On the 7473, CLR' = 1, J = K = 1, Q = 0. Give Q after three clock pulses. (b) Why do you see no change on the 7474 while you hold the pulser down? (c) Derive D for a T flip-flop from the D excitation ($D = Q_{n+1}$).

Answers

Opening question. Race-around: with J = K = 1 and a long CLK-high, the gate JK toggles repeatedly through its own feedback, so the final state is random. ICs trigger on a clock **edge** (7474 ↑, 7473 ↓), so they change once per pulse.

Check 1. (a) It toggles each falling edge: 0 → 1 → 0 → **1**. (b) The 7474 acts only on the **rising** edge, which happens when you press. Holding the button is a steady level, so nothing more happens. (A 7473 acts on the release, the falling edge.) (c) $Q_{n+1} = Q_n$ when T = 0 and Q_n' when T = 1, so **D = T ⊕ Q**.